



DIGITAL TRUST FUTURES
FOUNDATION

DIGITAL SOVEREIGNTY IN PRACTICE

The Cost of Illusory Sovereignty

Why false control creates national risk and loss of trust

Digital Sovereignty Series | Article 2 of 6

Why this matters now

False sovereignty is not just a policy weakness. In population-scale Digital Public Infrastructure, weak control can become an economic, privacy, service-delivery and trust crisis. This article explains what a country risks when it believes its digital foundations are under control before that control has been technically proven.

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Executive summary

Digital sovereignty is sometimes treated as an abstract policy preference: a matter of national ambition, technology positioning, procurement strategy or cloud-hosting choice. For countries deploying Digital Public Infrastructure at scale, this framing is too weak. Digital sovereignty is better understood as a practical national risk-control requirement, because identity systems, payment rails, health platforms, social protection systems and data exchanges now sit close to both citizens and the operating capacity of the state.

When control over these systems is weak, the consequences are not confined to an ICT department, a project team or a single ministry. They can affect national service continuity, citizens' rights, economic participation, fiscal integrity, public trust and national resilience simultaneously. This is why a false sense of sovereignty is dangerous: **it creates confidence before control has been proven.**

- A country may believe its data is sovereign because it is hosted locally.
- It may believe its platforms are independent because a domestic contractor has been appointed.
- It may believe it has avoided dependency by selecting open-source Digital Public Goods.

Each of these choices can support sovereignty, but none of them proves sovereignty on its own. The real test is whether the country can govern, operate, secure, verify and adapt its digital foundations under legal pressure, technical failure, vendor disruption, cyberattack or geopolitical stress.

What leaders should test before claiming sovereignty

- 1 Can the country keep the system running if the primary vendor, cloud provider or integration partner becomes unavailable?
- 2 Can the country prove who holds the encryption keys, who can access sensitive data, and under which legal authority access can be compelled?
- 3 Can domestic or trusted regional teams inspect, modify, recover and secure the system without waiting for a foreign supplier?
- 4 Can the country independently verify the security of the system beyond vendor assurance, implementation reports or compliance claims?
- 5 Can vulnerabilities in a shared Digital Public Goods component be tracked, patched and coordinated across deployments before they become systemic risks?

False comfort	Real exposure	Leadership question
"The data is hosted locally."	Local hosting may satisfy residency but not sovereignty if legal access, key custody or administrative control is elsewhere — for example where a foreign government can compel a global cloud provider under its own law.	Who can actually access the data, and who holds the keys?
"A domestic supplier was appointed."	A local contractor may still depend on foreign cloud infrastructure, tooling, updates, support or managed services.	Is the stack domestically controllable, or only locally controllable?

False comfort	Real exposure	Leadership question
"The platform is open source."	Open source reduces proprietary lock-in, but does not automatically solve hosting, data control, operations or supply chain risk.	Can we operate, patch and assure the platform independently?
"The system has passed compliance."	Compliance may show policy alignment, but not resilience under attack, outage, subpoena or vendor failure.	Has control been tested under realistic failure conditions?
"Citizens trust the programme."	Trust can collapse quickly after exclusion, fraud, surveillance concerns, data misuse or service disruption.	What evidence shows that security, safety and accountability are working in practice?

The first article in this series argued that real digital sovereignty starts with control, regardless of where the data is hosted or whether the software is open source. This second article shows why the absence of that control is not merely theoretical; it can lead to crises in privacy, service delivery, the economy, trust and national resilience.

1. Sovereignty failure is a material national risk

The absence of meaningful digital sovereignty is not a narrow ICT concern. When a country digitises identity, payments, health, social protection and data exchange, the systems involved become part of the country's public operating model. They determine how people prove who they are, how money moves, how benefits are delivered, how health information is accessed, and how agencies exchange data. A failure of control can therefore simultaneously affect service continuity, citizens' rights, national security, fiscal integrity and economic participation.

The central risk is false confidence, and it is especially dangerous because it often appears to be progress. A government may point to a local data centre, a domestic contractor, a cloud residency commitment, an open-source platform or a signed compliance report and assume that sovereignty has been achieved. In reality, each of these may represent only partial control: useful but incomplete.

Three practical risk tests

Every country should apply these before treating a national digital system as meaningfully sovereign.

- 1 Control under pressure.** Can the country keep the system running if a vendor, cloud provider, integration partner or external support team becomes unavailable, unwilling, legally restricted or commercially difficult?
- 2 Authority over sensitive data.** Can the country prove who holds the encryption keys, who can access personal and biometric data, what legal authority governs access, and whether any foreign jurisdiction can compel disclosure without effective domestic control?
- 3 Independent verification.** Can the country assess the security, resilience, supply chain and operational maturity of the system independently of the parties that built, host, fund or operate it?

If those claims are not supported by operational capability, key custody, independent assurance and resilience planning, the country may discover the gap only after a breach, subpoena, outage or geopolitical shock. That is why a failure of sovereignty must be treated as a material national risk. It is

not simply a technical weakness to be remediated later; it is a governance, security, privacy and continuity exposure that can quietly accumulate until the moment the country most needs the system to hold.

2. Strategic control and economic dependency

Strategic control is weakened when the roadmap, pricing, support model, security posture and continuity of national systems are effectively determined by actors outside the country's own institutions. The issue is not that all external vendors are unsafe, or that countries should reject international technology, cloud services, donor support or specialist implementation partners.

In many cases, these actors bring essential capability, scale and innovation that would be difficult or costly to reproduce domestically. The real issue is **dependency without leverage**: a condition in which a critical national platform continues to function only because an external provider remains commercially willing, legally able, politically permitted and technically available to support it.

- 1 Roadmap dependency.** When the country cannot prioritise critical changes, security improvements, localisation needs or integration requirements because the underlying platform evolves according to an external vendor's product roadmap.
- 2 Pricing dependency.** When the cost of running national systems is exposed to foreign currency pressure, licensing changes, consumption-based cloud pricing, or contractual renewal terms that the country has limited power to challenge.
- 3 Continuity dependency.** When an outage, sanctions exposure, legal dispute, support withdrawal, cyber incident or geopolitical shift affecting the provider becomes, in practice, a national service-delivery risk.

For smaller and lower-income countries, the bargaining challenge is especially difficult. Market size influences negotiating power, and a small country may need world-class infrastructure but lack the scale to demand bespoke sovereignty controls, stronger audit rights, local operational guarantees, external key management, exit provisions or transparent incident reporting terms on its own.

Why regional cooperation is a sovereignty instrument

Regional arrangements, common procurement standards, shared cloud and security assurance requirements, pooled technical expertise and joint negotiation mechanisms are not merely ideals of cooperation; they are practical instruments of sovereignty. By acting together, smaller countries can convert fragmented demand into collective leverage and reduce the risk that each country negotiates alone from a position of structural weakness.

Economic dependency also carries a longer-term strategic cost. When national systems depend heavily on externally controlled platforms without deliberate skills transfer, domestic operations, security capability and exit planning, the country may pay continuously for access without building durable internal competence. Over time, this can turn digital transformation into a recurring import bill rather than a foundation for local capability, public-sector resilience and the development of a regional digital industry.

The goal is not technological self-isolation. The goal is to ensure that external capability strengthens national control instead of replacing it. A country can use global technology wisely and still insist that

the final authority over continuity, data access, security assurance and system evolution remains in the accountable hands of domestic or trusted regional authorities.

3. Population-scale harm and PII exposure

Digital Public Infrastructure differs from ordinary enterprise technology because the blast radius of failure can be the entire population. A commercial platform may affect a defined customer base, a business unit or a market segment. DPI affects citizens across identity, payments, health, welfare, mobility, education, taxation and public entitlements. When these systems fail, the consequences are not only data loss or service disruption; they can also affect a person's ability to prove who they are, receive public support, access healthcare, move money, enrol in services or participate in the formal economy.

This is why personal information within DPI must be understood as **national risk data**, not merely compliance data.

DPI data category	Why it is sensitive	Sovereignty risk if control fails
National identity numbers and civil registration data	They anchor access to public and private services.	Exposure can enable fraud, impersonation, exclusion or long-term identity abuse.
Biometric identifiers	Fingerprints, iris patterns and facial templates are permanent or difficult to replace.	Once compromised, they cannot be meaningfully reissued like passwords or cards.
Payment and transaction data	They reveal financial behaviour, relationships, locations and economic participation.	Misuse can expose citizens to fraud, profiling, surveillance or financial exclusion.
Social protection and welfare records	They contain information on vulnerability, poverty, eligibility, household and dependency.	Exposure can stigmatise citizens, distort benefits or enable targeted abuse.
Health information	It contains highly sensitive personal, family and clinical data.	Breach or misuse can cause discrimination, loss of dignity, safety risks and irreversible damage to trust.
Consent and data-exchange records	They determine which agencies or providers can access which data, for what purpose, and when.	Weak controls can enable unauthorised sharing, replay, excessive access or misuse of purpose across systems.

The risk is also cumulative. A breach in one system can expose another because DPI components are frequently integrated by design:

- A digital identity credential may authorise a welfare payment.
- A consent flow may release financial or health information.
- A data-exchange layer may connect multiple agencies through shared APIs.

A vulnerability in one component can therefore move across the ecosystem, turning a single system weakness into a multi-platform exposure. This is especially important in countries that adopt

interoperable Digital Public Goods and shared, reusable DPI components. Interoperability is valuable because it reduces duplication and improves service delivery, but it also means that weak authentication, poor API governance, exposed credentials, misconfigured access controls or unpatched software dependencies can have consequences beyond the first affected system.

Population-scale harm is ultimately a trust and legitimacy issue. When citizens believe that digital systems expose them to identity theft, surveillance, fraud, exclusion or uncontrolled data sharing, they may avoid the very platforms designed to improve inclusion and public service delivery. The stronger sovereignty question is therefore not only whether the data exists, where it is hosted, or whether a privacy policy is in place; it is whether the country can prove that citizens' most sensitive information remains protected, controlled and accountable across the full DPI ecosystem.

4. Trust as a system user requirement

Communications campaigns, launch events or public messaging do not create trust. It is earned when security holds, citizens experience safety in practice, and the system behaves reliably when people need it most.

A Digital Public Infrastructure system can be technically innovative, interoperable and well-funded, but still fail its public purpose if people associate it with exclusion, surveillance, fraud, unauthorised access, poor recourse or service interruption.

This matters because adoption is the development case for DPI. Citizens do not adopt identity, payment, welfare, health or data-exchange systems only because they exist. They adopt them when they believe the system is safe, useful, fair and accountable. Trust therefore becomes a functional requirement, not a soft communications objective.

Trust is weakened when citizens experience or perceive:

- **Exclusion:** when authentication failures, poor data quality or weak fallback processes prevent people from accessing services.
- **Surveillance risk:** when citizens believe their identity, payment, welfare or health data may be used beyond its intended purpose.
- **Fraud and loss of recourse:** when payment users fear unauthorised transactions, identity misuse or unclear accountability after harm.
- **Service interruption:** when outages, cyber incidents or vendor failures prevent access to essential public services.
- **Unclear accountability:** when citizens cannot understand who is responsible, how errors are corrected, or how misuse is investigated.

If citizens do not trust a digital identity system, they may avoid enrolment or resist linking it to other services. If payment users fear fraud or lack confidence in dispute resolution, they may remain in cash. If vulnerable people are excluded from social support because of authentication, biometric, connectivity or data-quality failures, the programme's legitimacy is undermined, even if the architecture appears sophisticated.

Trust is therefore not an outcome that can be declared after deployment. It must be designed into security controls, privacy protections, user experience, fallback channels, grievance mechanisms and independent assurance from the beginning. In DPI, trust is the public evidence that sovereignty, security and safety are working together.

5. Shared DPG ecosystems create shared risk

Open-source Digital Public Goods create enormous value by allowing countries to reuse proven components, accelerate implementation, reduce duplication, improve interoperability and avoid long-term proprietary lock-in. For many countries, especially those with limited budgets and urgent service-delivery needs, DPGs offer a practical path to building national-scale digital systems without having to start from scratch.

But the same reuse model that makes Digital Public Goods powerful also creates systemic exposure. A vulnerability in a widely adopted component, dependency, API pattern, deployment script, configuration template or integration module may be present across many national deployments simultaneously. If countries lack mature software supply chain monitoring, patch governance, vulnerability disclosure processes, version tracking and independent security testing, a weakness discovered in one deployment can quietly remain active in another.

- 1 Common-code risk:** the same platform, library or dependency may be reused across multiple countries, creating a shared technical weakness if a vulnerability is discovered.
- 2 Common-configuration risk:** countries may copy similar reference architectures, deployment guides, API patterns or integration models, repeating the same misconfiguration at scale.
- 3 Common-response risk:** if patching, disclosure and version management are not coordinated, one country may fix a weakness while another remains exposed for months.

This is not an argument against Digital Public Goods

It is an argument for a more honest model of shared responsibility across the entire ecosystem. Countries adopting shared platforms should maintain software bills of materials, vulnerability disclosure processes, version tracking, independent code and architecture review, secure configuration baselines and regional coordination mechanisms.

Funders and ecosystem stewards also have a role to play by supporting security maintenance, coordinated assurance, responsible disclosure channels and common minimum security expectations across deployments.

The sovereignty lesson is clear: open source reduces one form of dependency, but it does not remove the need for disciplined security governance and effective DevSecOps assurance practice. A Digital Public Good becomes safer when every country using it contributes to the health of the shared ecosystem, not when each deployment treats security as a private national matter. In a connected DPI environment, one country's unpatched weakness can become another country's inherited exposure, and shared infrastructure demands shared vigilance.

Series bridge

The next article turns to the threat environment that makes these sovereignty risks urgent: cyberwar. It explains why Digital Public Infrastructure can no longer be treated as ordinary digital government infrastructure, but as critical national infrastructure operating in an active, contested and persistent threat environment in which identity, payment, health, welfare and data-exchange systems are strategic targets.

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Godfrey Kutumela is a globally recognised cybersecurity practitioner with over 26 years of experience shaping secure digital ecosystems across fintech, identity and public infrastructure environments. He operates at the intersection of Digital Public Infrastructure (DPI), Digital Public Goods (DPGs) and national-scale platforms, bringing deep expertise in securing systems where the impact of failure is measured at the population level rather than the enterprise level.

- Trusted advisor to governments, foundations and ecosystem builders delivering secure-by-design DPI across Africa, Latin America and global markets.
- Core contributor to large-scale platforms including Mowali (mobile money interoperability infrastructure), Mojaloop (interoperable faster payments), MOSIP (digital identity) and Tazama (fraud systems).
- Designs and embeds independent, risk-based security practices across payment systems, identity frameworks and cross-border financial infrastructure.
- Works across policy, architecture and implementation layers, translating global standards into deployable security controls.
- Contributor to the Digital Public Goods Alliance (DPGA) Privacy Expert Group, shaping global privacy and security indicators.
- Active in open-source security, DevSecOps integration and supply chain assurance for reusable digital platforms.
- Delivered independent security reviews, penetration testing and architecture governance across critical DPI implementations.

His leadership philosophy in the DPI ecosystem is clear: security forms the foundation upon which safety and trust are built. Safety is what citizens experience in practice, and trust is the legitimacy that sustains adoption; once lost, it is difficult to restore.

About Digital Trust Futures Foundation

Digital Trust Futures Foundation is an independent, non-profit public-benefit organisation strengthening the safety, security, rights, inclusion and trustworthiness of digital public infrastructure (DPI), digital

public goods (DPGs), open-source technology and AI-enabled public services. Its work is grounded in the recognition that DPI has become critical national infrastructure operating in a persistent threat environment, requiring independent, ecosystem-level assurance that goes beyond traditional approaches.

The Foundation operates outside vendor, implementer and platform incentives. It combines evidence-based cybersecurity, safety and AI-governance assurance with human-rights and child-protection expertise, and releases its tools, research and frameworks as reusable public goods rather than proprietary products — so that the value of a single engagement extends to every government, maintainer and community facing the same risk.

Digital Trust Futures Foundation NPC is registered as a non-profit company in South Africa and works across Africa and the wider Global South, with emphasis on countries and communities adopting DPI under capacity and assurance constraints.

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